

Category: Implementation & Operations in Mininet

Scenario 9

A student is tasked with creating a simple network topology for a lab assignment: one switch connected to four hosts.

Question: What single Mininet command-line instruction would create this topology and verify that all hosts can communicate with each other?

Answer:

To create a single switch connected to four hosts and immediately test connectivity between all hosts in a single command, the student would run:

```
sudo mn --topo single,4 --test pingall
```

Explanation:

- `sudo mn` : Starts Mininet with root privileges.
- `--topo single,4` : Constructs a topology consisting of one single switch connected to 4 hosts (h1, h2, h3, and h4).
- `--test pingall` : Automatically executes an all-to-all ICMP ping test between every pair of hosts to verify full connectivity, and then exits Mininet once complete.